GENTAMICIN

Nephrotoxic & ototoxic

Therapeutic drug monitoring

CHEMICAL NAME Mixture of gentamicin C1, C1a and C2 aminoglycosides (as sulfate) from *Micromonospora purpurea*

CLASS Aminoglycoside antibiotic

TRADE NAMES Garamycin · Genticyn · Cidomycin · Gentamicin sulfate injection

THERAPEUTIC USES / INDICATIONS

- Serious Gram-negative infections: sepsis, pyelonephritis, intra-abdominal infection.
- Neonatal sepsis (with ampicillin/benzylpenicillin).
- Endocarditis (synergy with penicillin/vancomycin).
- Topical: eye and ear drops, skin cream.

ADMINISTRATION

- **IV** infusion over 30–60 min (extended-interval dosing) or slow IV over 3–5 min; **IM**.
- Do not mix with penicillins or cephalosporins in syringe/line.
- Dose on ideal or adjusted body weight in obesity.
- Take levels as per local protocol (e.g. 6–14 h after dose for once-daily regimen; trough before next dose).

DOSAGES

- **Once-daily (adult):** 5–7 mg/kg IV once daily; interval adjusted to levels and renal function.
- **Multiple daily:** 1–1.7 mg/kg every 8 h.
- **Endocarditis synergy:** 1 mg/kg every 8–12 h (low-dose).
- **Neonates:** 4–5 mg/kg every 24–36 h (age-dependent).
- Target trough <1 mg/L (once daily).

PHARMACOKINETICS

- **Onset:** IV immediate. Poor oral absorption; poor CSF penetration.
- Not metabolised.
- Excreted unchanged by glomerular filtration. **t½:** 2–3 h (much longer in renal impairment, neonates).

PHARMACODYNAMICS

- Binds **30S ribosomal subunit** → misreading of mRNA → bactericidal.
- Concentration-dependent killing with post-antibiotic effect (basis of once-daily dosing).
- Active against aerobic Gram-negatives incl. *Pseudomonas*; synergy with β-lactams against enterococci/staph.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain.
- Nausea, rash.

ADVERSE EFFECTS (SERIOUS)

- Nephrotoxicity (acute tubular necrosis; usually reversible).
- Ototoxicity (vestibular and cochlear — often irreversible): tinnitus, vertigo, hearing loss.
- Neuromuscular blockade → respiratory paralysis (with anaesthetics, myasthenia).
- Hypomagnesaemia.

CONTRAINDICATIONS & CAUTIONS

- Myasthenia gravis.
- Hypersensitivity to aminoglycosides.
- **Caution:** renal impairment, elderly, pregnancy (fetal ototoxicity), pre-existing hearing loss, mitochondrial gene mutation (m.1555A>G).

KEY INTERACTIONS

- **Loop diuretics (furosemide):** ↑ ototoxicity.
- **Vancomycin, ciclosporin, amphotericin, cisplatin, NSAIDs:** ↑ nephrotoxicity.
- **Neuromuscular blockers:** prolonged paralysis.
- **Penicillins:** in-vitro inactivation.

NURSING MANAGEMENT

- Weigh patient; confirm dosing weight and renal function.
- Time the infusion and blood levels accurately and document.
- Monitor creatinine, urine output, levels daily.
- Ask about tinnitus, dizziness, hearing changes.
- Ensure adequate hydration.

PATIENT EDUCATION

- Report ringing in the ears, dizziness, hearing loss or reduced urine.
- Drink enough fluids unless restricted.

KEY POINTS

- Aminoglycoside — Gram-negative, bactericidal.
- Nephro- and ototoxic.
- Once-daily 5–7 mg/kg; monitor levels.
- Avoid with furosemide, vancomycin when possible; never in myasthenia.

SOURCES CHECKED **[163]** Gentamicin 1 mg/ml solution for infusion (UK SmPC) **[164]** Gentamicin, IM, IV (US label)
[165] Gentamicin, IM, IV (US label, Hikma Pharmaceuticals USA Inc)

Full references p. 432–457